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RESEARCH

Solitary wave expansion and impact decimation in short inverse tapered granular chains

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Abstract
We present a simple, strongly nonlinear and short inverse tapered granular alignment (radii small to big) that can transform an impulse into a progressively wider and slower moving solitary wave. The system curiously also serves as an impact dispersion system for an impulse initiated at the smallest grain.

Keywords Granular Chain, Tapered Chain, Solitary wave, Impact absorption

1 Introduction

The Fermi-Pasta-Ulam-Tsingou (FPUT) problem [1–4] on the dynamics of masses connected by springs represented by both linear and weakly nonlinear potentials has deeply influenced the study of nonlinear dynamics through the past 70 years [4, 5]. In a similar vein, the 1983 and later studies of how a striker impulse effected at one end of an alignment of monodispersed elastic spheres, e.g., stainless steel spheres, propagates down the chain studied by Nesterenko [6–17] has had a profound impact on the theoretical, simulational and also on experimental progress on the effects of strong nonlinearity in discrete as well as in continuous systems. Further, both the systems admit the propagation of stable solitary waves (SWs) with slightly different properties [18–24]. For example, the width of the SWs [22–24], however measured, is different in each case. In both cases, the SWs interact with the boundaries and with one another and eventually such interactions lead to changes in the original waves and the production of what is often called secondary SWs [11], which are very low energy SWs that move slowly and are produced in the vicinity of the collision regions. Consequently, due to the production of these secondary SWs, the initial perturbation imparted to the system eventually disperses at late enough times such that the system slowly slides into an equilibrium-like state except that, in general, energy equipartition does not hold in this late time equilibrium-like state. This state has been referred to as the quasi-equilibrium state [25–28].

A key difference between the FPUT problem and the granular (i.e., Nesterenko's) system, which we shall call a granular chain, is that in the latter grain-grain contact can easily break resulting in abrupt cessation of the grain-grain interaction. Such a possibility can never arise in the FPUT system. When two adjacent grains are squeezed towards each other, only a compression pulse forms in the granular chain. In the FPUT system such an event would be associated with the formation of a compression and a dilation pulse, which would lead to a SW-anti SW pair [24].

Just like the FPUT problem continues to intrigue nonlinear dynamicists seven decades after its initial appearance [1, 29, 30], so does the granular chain problem, except it is much more recent. It is also somewhat simpler, but still presumably with many properties that remain unknown. In fact, one might even argue that the granular chain and the chain-inspired systems provide a simpler and rich playing ground for exploring the fundamentals and applications of an intrinsically nonlinear, non-integrable system (see for example in [31]). In the past four decades, significant progress has been made in the study of the dynamical behavior seen in granular chains [15] but a great deal remains to be explored still. It has turned out that these systems possess

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rich nonlinear dynamics, and many technologically significant applications [32–43]. These applications include systems that are potentially capable of converting noise into useful energy, tapered chain impact dispersion systems, systems designed for acoustic insulation, and inverse tapered chains which can be used for significant amplification of low frequency acoustic (e.g., infrasonic) signals and more.

In the present study, we consider the recently introduced inverse tapered chain (ITC) system held between fixed, rigid end walls where the grain radii grow in size progressively from one direction to another according to a fixed growth factor, say left to right (see Fig. 1) [42, 43]. In the ITC, the smallest grain is perturbed as opposed to the TC where the largest grain is perturbed. The TC systems have been extensively examined in the literature and are well known for their impact decimation properties, which we shall return to in Sect. 3.2 below.

We assume that all the grains are made of the same material (say stainless steel). By integrating the dynamical equations of motion for our system we show that the ITC system develops sustained grain-grain compressions toward the large grain end, which result in an unexpected time-dependent increase in the width of the propagating SW-like object that moves through the system upon the initiation of an impulse. We also show that the system is capable of impact dispersion. The increasing SW width behavior potentially raised a possibility of novel shock speed slowdown and shock dispersion across short length scales. We are especially interested in small system sizes in view of the fact that long ITC systems would be difficult to realize in the laboratory.

2 Model

Let us consider a chain of elastic spherical grains where the grains are held between rigid end walls and are in intimate contact at time $t = 0$. We assume that the grain radii progressively increase by some factor q as one travels along the chain from the small grain end (say, the left end) to the opposite end (say, the right end). The grains interact via the Hertz potential [44, 45]. We can describe the potential energy associated with the compression between the adjacent grains i and $i + 1$ of radii R_i and R_{i+1} as follows: let us define what is called *overlap* between the two adjacent grains as $\delta_{i,i+1}$ where $\delta_{i,i+1}(t) \equiv R_i + R_{i+1} - z_{i,i+1}(t)$ where $z_{i,i+1}(t) \equiv z_{i+1}(t) - z_i(t)$ represent the actual distance between the centers of the two adjacent spheres measured from some chosen origin for z. Observe that due to the geometry of the system it would not be possible to impose constant precompression on each of these grains. However, position dependent precompression could still be achieved. For the sake of simplicity, we ignore that case here. So one can see that in the absence of such precompression and any perturbation, $\delta_{i,i+1}(t = 0) = 0$. Finally, once a perturbation has been initiated, dynamic compression between grains i and $i + 1$ will end up being $R_i + R_{i+1} - (z_{i+1}(t) - z_i(t))$, i.e., the overlap $\delta_{i,i+1}(t) > 0$. Thus, $\delta_{i,i+1}(t)$ gives the dynamic compression shared between grains i and $i + 1$. The interaction energy between the spheres is expressed as a function of this overlap as follows:

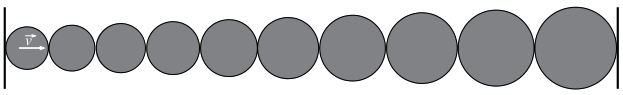


Fig. 1 A sketch of an inverse tapered granular chain showing the initiation of the impulse on the smallest grain on the left is shown. The system is assumed to be cased in a container (not drawn) with frictionless walls

$$V(\delta_{i,i+1}) = \frac{2}{5D}\sqrt{\frac{R_i R_{i,i+1}}{R_i + R_{i+1}}}\delta_{i,i+1}^{5/2} \equiv a_{i,i+1}\delta_{i,i+1}^{5/2}, \tag{1}$$

where the constant

$$a_{i,i+1} \equiv \frac{2}{5D}\sqrt{\frac{R_i R_{i,i+1}}{R_i + R_{i+1}}}, \tag{2}$$

and where

$$D_{i,i+1} \equiv \frac{3}{4}\left(\frac{1-\sigma_i^2}{E_i} + \frac{1-\sigma_{i+1}^2}{E_{i+1}}\right), \tag{3}$$

in which σ_i and σ_{i+1} and E_i, E_{i+1} describe the Poisson's ratios and the Young's moduli of the grains i and $i + 1$, respectively. We will consider all grains to be made of the same material and therefore replace $D_{i,i+1}$ by D here. Obviously, $V = 0$ if $\delta_{i,i+1} \leq 0$, i.e., if the grains are not in intimate contact. The above law is originally due to Hertz and is often referred to as the Hertz law [44]. The power of δ in Eq (1) is related to the nature of the overlap between the grains. For spherical grains, the overlap regions would in general be elliptical. Thus, it is possible to change the power on δ by engineering grains of different shapes as discussed by Sun et al. [45]. The progressive increase in the grain radii is described as

$$R_{i+1} = R_i(1 + q), \tag{4}$$

where typically $q \ll 1$. For $q = 0$ we recover the monodispersed chain, discussed above [46, 47].

For our work where there is no precompression, the dynamical equations of motion can be written as

$$m_i \ddot{z}_i = \frac{5}{2}(a_{i-1,i}\delta_{i-1,i}^{3/2} - a_{i,i+1}\delta_{i,i+1}^{3/2}). \quad (5)$$

Observe here that the right hand side of Eq 5 in the absence of any precompression is intrinsically nonlinear. Hence, unless the system is precompressed, no sound propagation is admissible here. This hence is a "sonic vacuum" system as referred to in Nesterenko's work [6, 8].

Since we are dealing with macroscopic grains, there inevitably is energy loss to the internal degrees of freedom of the grains. While there are many ways to model such restitutional losses, a particularly simple and well tested approach is due to Walton and Braun [48, 49]. In the Walton-Braun approach, the work done by the loading force during the loading of two adjacent grains exceeds the work done due to the unloading process. Hence, the difference between the force-displacement plots of the work done due to the two forces ends up revealing the energy lost to the internal degrees of freedom. Thus, following Walton and Braun [48] we write the restitutional loss w as follows:

$$\frac{F_{\text{unloading}}}{F_{\text{loading}}} \equiv 1 - w, \quad (6)$$

where F_{loading} represents the loading force between two adjacent grains as they press against each other and $F_{\text{unloading}}$ is the unloading force between the grains as they come apart. If the two forces turn out to be exactly the same all along the loading and unloading processes, no energy is lost in the work done by the forces during the process and $w = 0$ and hence total mechanical energy must be conserved and a perfectly elastic process must be realized. The restitution coefficient w defined above is not well known for most materials. We hence treat w as a free parameter and explore system dynamics for reasonable values of w. For the work presented in Sect. 3.1, we set $w = 0$.

To make sure our model system can be intuitively related to, we have used the following system parameters in all our dynamical simulations. The grains are assumed to be titanium grade 5 (Ti6Al4V) [50] with density (needed to find m_i) $\rho = 4.420$ mg/mm^3, $E = 113.18$ N/mm^2, $\sigma = 0.3$, and the calculated $D = 0.01206$ mm^2/N. The size of the smallest grain $R_1 = 5.0$ mm. The system is held between rigid end walls (Fig. 1). The dynamics is initiated with a velocity $v_0 = 1.0$ m/s. We have used a velocity-Verlet predictor-corrector integration algorithm [51] for our simulations with an integration time step $dt = 1 \times 10^{-5}\mu$sec and the runs are made across $2 \times 10^8 - 5 \times 10^8$ steps. This turns out to be between 2 and 5×10^{-3} s. In the absence of dissipation, i.e., when $w = 0$, energy conservation was perfect to 8 decimal places across the entire set of runs.

In our current study we will consider ITC systems with $q = 0.05,\ 0.07,\ 0.09,\ 0.11,\ 0.13$ and some 0.15 cases. The number N of grains in the system will typically be between 10 and 20 in our studies. Larger systems are likely not easily realizable in the laboratory. The values of $w = 0,\ 0.1,\ 0.3,\ 0.5$ and 0.7 have been used in our studies. To our knowledge, dissipation larger than 0.5 is not common in typical systems.

3 Results

Recently, the ITC system was shown to be capable of filtering out high frequency sound waves and capable of amplifying low frequency sound waves, including at frequencies down to the infrasonic range [41–43]. However, the basic physics of *impulse transmission* in this system, perhaps the most fundamental dynamical problem that can be probed [6, 18, 19], remains to be understood. As mentioned above, there are two key aspects of the impulse propagation problem in the ITC that we focus upon here. In Sect 3.1, we focus on the new phenomenon of SW expansion in this system, emphasizing that the SW is an approximate SW compared to the usual perfect SW encountered for a monodispersed granular alignment [52]. In Sect. 3.2, we discuss the ITC as an energy dispersion device. This is not the strongest point of the ITC. As we shall see, the TC is a better impact dispersion system. However, the ITCs impact dispersion features of soft and highly energy absorptive grains (with high restitutions) might nevertheless be of interest in engineering applications.

3.1 SW expansion

In this section we will present the problem of the expansion of a propagating SW in an ITC, which is held between rigid end walls. While we have done extensive simulations for different system sizes denoted by N, different tapering parameters q, restitution w values and a range of initial velocities $v_0(t = 0)$, we present our dynamical simulation study of the system using the results for the $N = 15, q = 13$ and $v_0 = 1.00$ m/s. We reiterate that the total energy was perfectly conserved in our studies for $w = 0$ cases.

As shown in Fig. 1, we assume that the ITC is held between rigid end walls. All the grains are at rest in their equilibrium positions, each gently touching the neighbor on both sides but with zero overlap. Dynamics is initiated in the system via a velocity impulse $v_0(t = 0)$ initiated at the smallest, i.e., the leftmost grain. Let us call this the first grain. This grain is driven in the rightward direction, i.e., into the neighboring grain. The entire energy of the system then becomes the kinetic energy of this first grain until it

starts compressing against the second grain, which leads to the transfer of kinetic energy to potential energy and a subsequent decompression phase of the first grain as it reverses course, etc.

As for the collisions, this first grain collides with the slightly larger second grain, to whom it gives most of its energy, which then immediately starts compressing the third grain and the third the fourth and so on. Soon after the recoil of the first grain toward the back wall, the second grain moves toward the first grain and so on. In each forward collision, most of the energy in a grain is transferred forward. The recoil and subsequent low velocity collisions suffered by the grains towards the left end culminate in repeated collisions and highly complex dynamics at the smaller grain end of the ITC, while most of the energy of the impulse moves forward. We assume the walls in which the grains are kept to be frictionless. Hence only the translational motion of the grains have been considered in this study (see Fig. 1).

Assuming for simplicity that we are dealing with hard spheres, it is easy to use momentum and energy conservation to see that recoil of the smaller grain would be of zero velocity if the masses were the same and equal in magnitude. And the recoil velocity of the first mass would equal the magnitude of the initial velocity if the larger mass were infinitely massive or a rigid wall. If we now impose $q \ll 1$, we observe that the masses grow very slowly with increasing grain number and thus most of the energy will be transferred forward. Further, as discussed in Ref. [53], the SW that is propagating through the system is nominally 3 grain diameters wide [23] with some 80% of the kinetic energy of the SW sitting on one grain at the geometric center of it [53]. Hence, gentle growth of masses and the subsequent widening of the grain sizes along the chain leads to only modest backscatter of energy. The progressive increase in grain sizes result in a gradual expansion of the SW with a small attendant reduction in the total energy carried by the SW. Thus, what we see in the simulations of the actual system for $N = 15$ grains, $w = 0$ and $q = 0.13$ in Fig. 2a is indeed what one might expect to see. Figure 2b depicts the same data but instead of grain number the actual distance from the origin has been used. Thus, Fig. 2(b) reveals that the energy of the excitation progresses through the system with only a slight slow down and not as dramatic a slow down as suggested in Fig. 2a.

The simulated propagating SW obtained via numerical integration of Equation 5 can be seen to quickly develop from a δ function impulse in velocity at the left most grain in Fig. 2 into a SW pulse by the time it reaches the third grain (see Fig. 2a and b). Though diminishing in energy ever so slightly as it progresses, as the SW moves into the ITC, the grains become progressively wider. Thus, the ITC shows the property that an impulse initiated into the system develops

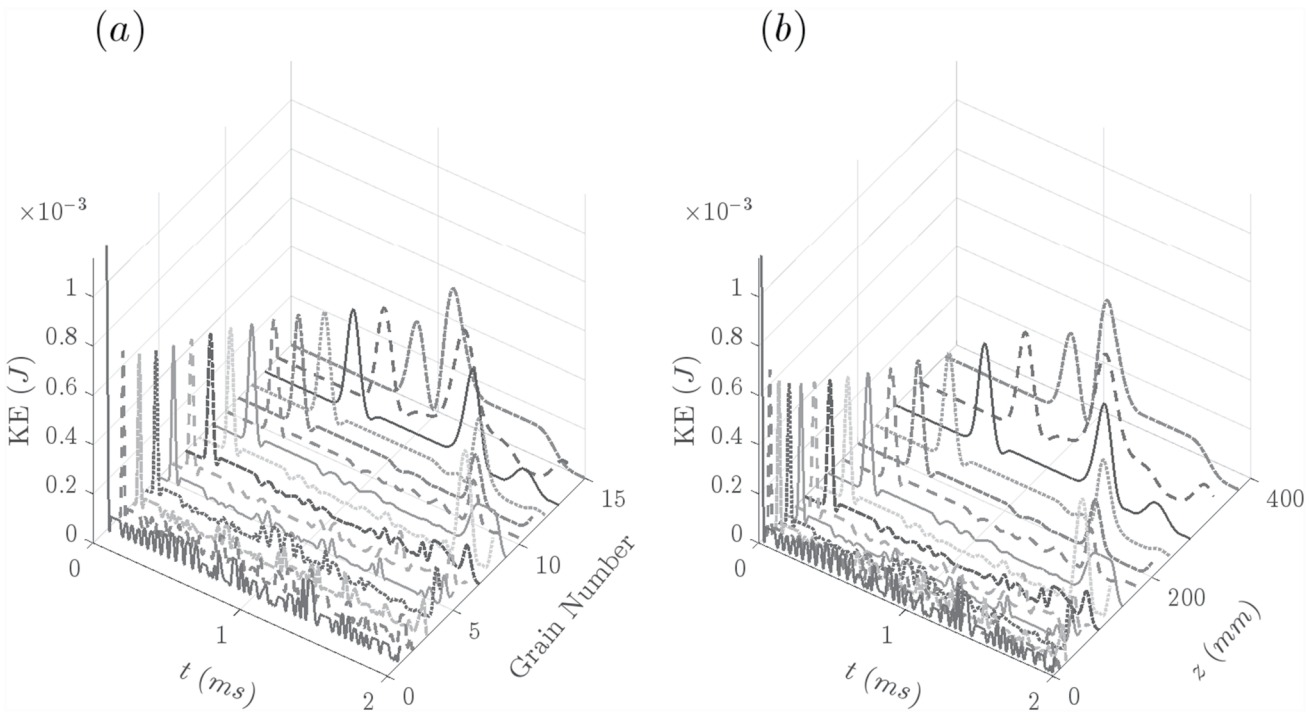


Fig. 2 (**a**) We show the kinetic energy of each particle denoted by a grain number as a function of time. (**b**) The same data as in (a) is shown except that the y axis has been replaced by the actual distance traveled by the perturbation measured from the left end of the chain. Observe that the slow down of the SW velocity in time is exaggerated when shown in terms of the grain number and the speed is very nearly linear when shown via actual distance traveled

into a SW, which remains stable as it moves toward progressively larger grains but becomes wider and wider in size. To our knowledge, this is the first study that shows that it may be possible to take a SW and find a way to expand it such that its energy is spread out over a larger space. Earlier work has shown that the width of a SW can be controlled by changing the nonlinearity in the force law [23] that describes the grain-grain contacts. The current study reveals a different way to broaden the waves by simply slowly enlarging the grain sizes.

The late time behavior of the grains in the ITC system reveals that the first 5 or 6 grains undergo a great many high frequency collisions with their neighbors. As the grains become larger and progressively slow moving, it takes longer for collisions between the larger grains to develop. Thus the collision frequencies drop substantially from one grain to another. The net effect of reduced collision is a smoother transfer of the energy forward. This is clearly visible in Fig. 2(b) where we see essentially unchanged *KE* after the passage of the SW by grain number 10.

Figure 3 describes the velocity versus time data of each grain for the system with $N = 15$, $w = 0$ and $q = 0.13$ system with $v_0 = 1.0$ m/s in the period up to 2 ms. As evident from Fig. 2, the SW collision with the back wall happens shortly before 1 ms. Figure 3 shows that some of

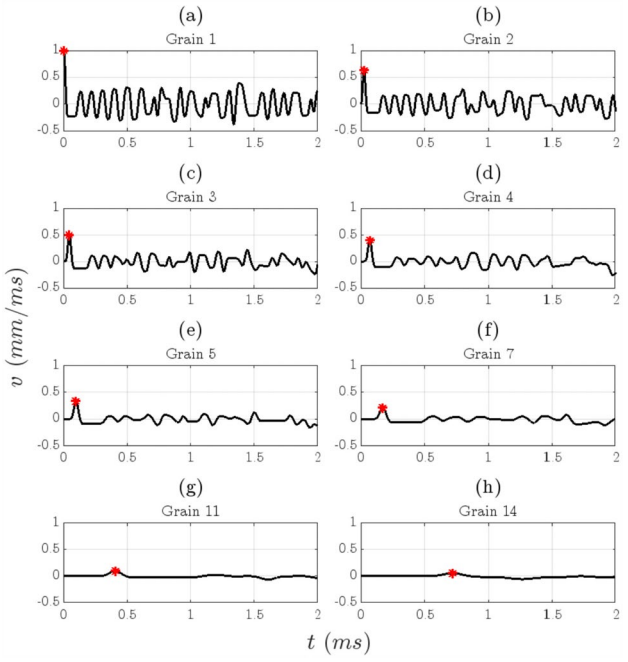


Fig. 3 We show the velocity (m/s) versus time (in ms) data for the grains in a 15 grain ITC with $q = 0.13$. Observe that a part of the energy resides for extended times in the tapered left end whereas velocities are only about 10% of the initial velocity in the right end where the larger grains reside. The small bumps in the velocity versus time plots show propagating SW structures are visible for grains 11 and 14

the system's kinetic energy remains in the smaller grains at the left end where the perturbation was initiated (see panels (a–e). The grains have very small velocities and hence relatively reduced dynamical activity towards the right end. To this end, observe that the grain velocities at the right end are less than 10% of the same at the left end.The dramatic slow down of the wave is not so evident from Fig. 2 when viewed through the lens of *KE*.

It is interesting that the dynamics of the larger grains on the right represent the typical traces of a passing SW, as confirmed by Fig. 2 and panels (g) and (h) of Fig. 3. When we look at the kinetic energy data of the grains across time however, as in Fig. 4, the energy transport picture looks quite different. It is clear that the high frequency dynamics of the smaller grains is substantially obliterated in the panels (a)–(o) when one looks at the grain kinetic energies. The increase in grain sizes leads the SW to slowly become wider as it propagates down the chain. In Fig. 4 we show grain kinetic energies at the specific times when the SW happens to pass through the centers of each of the grains. Given the radius of the grains increase linearly, this widening is also linear in the grain number. Further, observe that the maximum kinetic energy carried by a SW also slowly decreases linearly in time. The total area in a SW however turns out to be slowly decreasing as the SW propagates deeper into the ITC. This is expected in view of the backscattering that inevitably happens at each contact, no matter how weak the backscattered energy at each contact turns out to be. We conjecture that given energy conservation, the progressively smaller grain velocities and weaker backscattering at the contacts as the grains become larger is likely the ultimate reason why the ITC turns out to be a signal transmitter and an amplifier in the infrasound regime [43]. It is noteworthy that in the spatial and energy scale resolution used in Fig. 4, the width of the SW in all cases remains a single grain diameter wide.

It is worth noting that if the system is allowed to evolve for extended time scales, such as more than one complete roundtrip of the pulse along the ITC, repeated grain-grain collisions will spread throughout the chain, and the system will at late times likely slip into a near quasi-equilibrium-like phase [25] with an effective slight temperature gradient from being slightly hotter at the left to slightly cooler on the right end. This expected outcome is suggested from the data in Fig. 2. We have left aside performing such late time studies for future work because of the extensive computations needed and the ever-present possibility of error build up due to the large mass mismatches in the system.

We note though that given the low precision in space regarding the shape of the SW, we are unable to confirm whether the wave energy is best described by a $\mathrm{sech}^4 f(z - z_0)$ type function where $f(z - z_0)$ represents

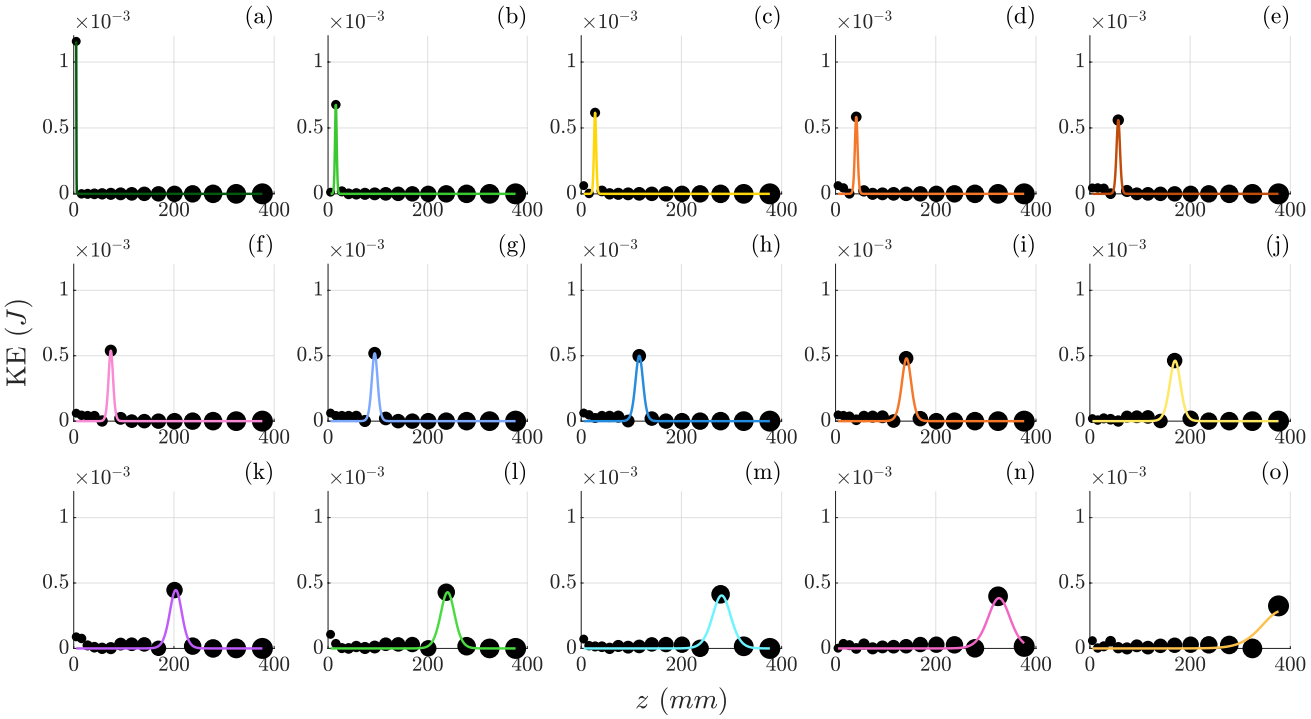


Fig. 4 Here we plot the kinetic energies of the individual grains at the right times so as to show when the SW is sitting on a grain of some size. As the perturbation proceeds to the right, the SW moves to progressively larger grains as is evident from the panels in the figure. Observe that though small, there is always some kinetic energy in the grains at the left end of the chain

an odd powered polynomial [52]. The description is consistent with the SW in the monodispersed system studied earlier.

We next look into the nature of the potential energies of the grains in the system in Fig. 5. To do this, we calculate the distance by which the grains are compressed with respect to their original equilibrium positions. Thus, the y axis parameter in Fig. 5, $\Delta z \equiv z_j - z_j^0$, where $z_j^0 \equiv \sum_{i=1}^{j-1} 2R_i + R_j$ for each grain $j \geq 1$, where the RHS $>(<)0$ means the grain has moved to the right (left) of the equilibrium position. Figure 5 shows that the system begins to rebound around the 0.9–1 ms mark and slowly decompresses with the grains moving to the left thereafter until in the last panel (p).

3.2 Impact dispersion in the ITC

The ITC is the backward version of the TC. As alluded to earlier, in the TC we studied how an impulse is broken down in a chain where the grain sizes shrink progressively along the direction of travel of the perturbation. In an ITC on the other hand, we examine the fate of an impulse when it is effected at the smallest grain and moves down grains of progressively larger radii. Not surprisingly, the two are completely different dynamical problems. Nevertheless, we ask here whether this system also is an effective impact dispersion system simply because that possibility needs to be investigated. We then ask, if indeed the ITC also is an impact dispersion system then which one is better? The TC or the ITC?

If we assume the grains are hard spheres and impose momentum and energy conservation, following the work in [39] we see that

$$v_{i+1}/v_i = 2/(1+\epsilon^3), \tag{7}$$

where $\epsilon \equiv (1+q)$ and $KE_{i+1}/KE_i = 4\epsilon^3/(1+\epsilon^3)^2$.

For N grains if there are $N-1$ collisions, we find that the normalized kinetic energy

$$KE_{\rm i} = (4(1+q)^3/[1+(1+q)^3]^2)^{i-1}, \tag{8}$$

which describes the normalized kinetic energy of the propagating SW under the hard sphere assumption at some grain i, where $1 \leq i \leq N$. $KE_{\rm i}$ in Eq 8 is obviously incorrect because it badly overestimates the kinetic energy of the propagating SW in the actual compressible system for $i > 1$. As outlined in Ref. [39], it is possible to develop an analytic expression for the normalized kinetic energy at the right end in the presence of energy loss. However, it is not easy to simply relate the energy loss for each grain to the restitution w used in our work. For this reason we refrain

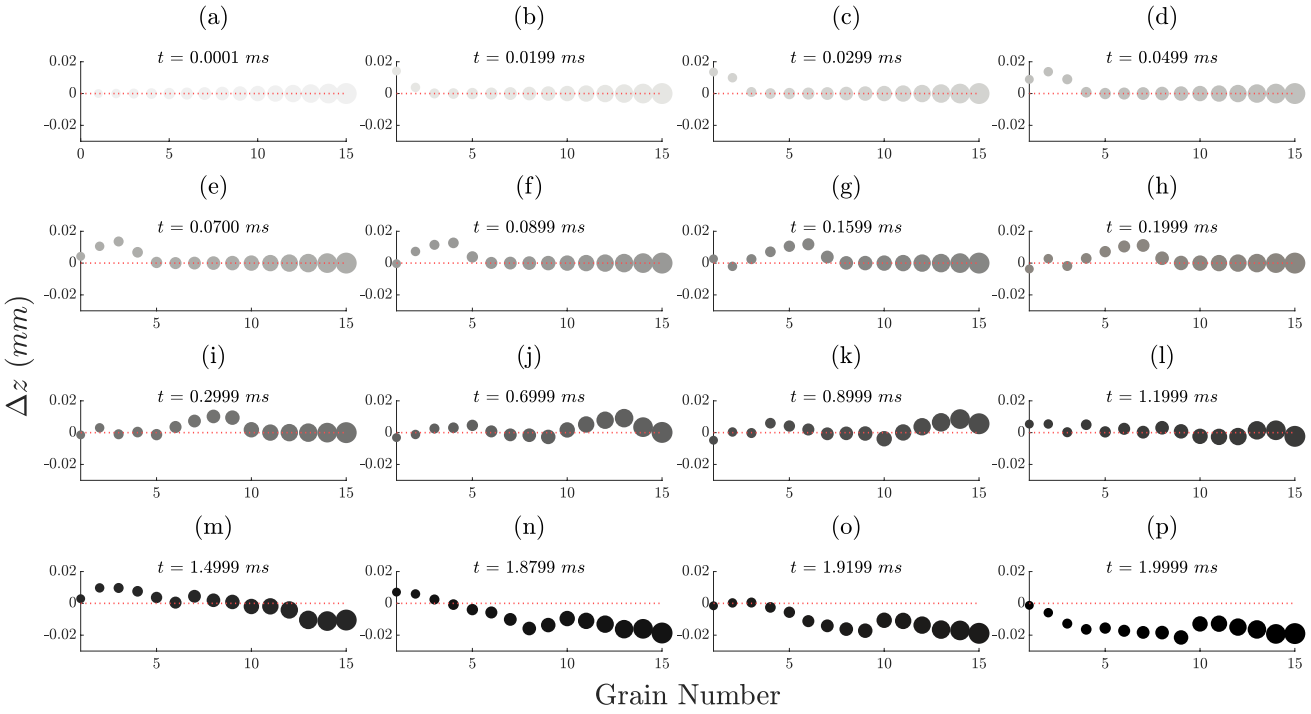


Fig. 5 The displacements for grain i are shown in panels (**a**) through (**p**), where a positive (negative) change in the equilibrium position Δz equates to a forward moving or rightward progression of the impulse. Panels (**b**) through (**e**) demonstrate the movement of the compression pulse until about the rebound of grain N against the end wall in panels (**j**) and (**k**), resulting in a phase of decompression by the larger grains as seen in panels (**n**) to (**p**)

from getting into a discussion of a hard sphere approximation in the presence of w.

The results of our simulation based studies are summarized in three "phase diagrams" that try to capture the behavior of the ratio of the maximum kinetic energy of the last grain with respect to the maximum kinetic energy of the first grain during the first sojourn of the perturbation. We call this quantity $KE_{\rm norm}$. In Fig. 6a we show $KE_{\rm norm}$ for $w = 0$ (hence an energy conserved system) as a function of the tapering parameter q and the system size (N) as described by Equation 8 (upper surface) and as described by the simulations (lower surface). In Fig. 6b we have captured our studies of $KE_{\rm norm}$ as a function of w and N for $q = 0.13$, i.e., 13% tapering, also as described by the simulations. In Fig. 6b we specifically focused on systems where the restitution is substantial. The numbers used for w would be appropriate for studies of soft and easily deformable materials where $F_{\rm unloading} \ll F_{\rm loading}$.

In the hard-sphere model for the system, the entire energy of the system is kinetic at all times. Thus, it is not surprising to see that in Fig. 6a, $KE_{\rm norm}$ at $N = 10$ is nearly 0.9 for $q = 0.05$. For the simulated system, in earlier work we have argued that for the monodispersed system, the kinetic energy of the propagating SW varies between 0.55 and 0.57 whereas the potential energy of the system varies between 0.43 and 0.45 [11, 25]. Thus, there is a sharp drop in the $KE_{\rm norm}$ of the system as the effects of the softness of the potential kick in. The role of the potential energy of these grains and the size of the SW-like structure are significant enough such that in Fig. 6a, $KE_{\rm norm}$ is about 0.38 for $N = 10$ and $q = 0.05$.

In Fig. 6a, given the very limited range of values of q, we are unable to unequivocally determine how $KE_{\rm norm}$ depends on q. The number of data points available to study KE_{norm} versus w is only 4. This is why we are unable to unequivocally determine its behavior. The behavior of KE_{norm} versus w with the few data points is consistent with an exponential decay, however. Our analysis on the decay of $KE_{\rm norm}$ versus N in Fig. 6b shows that the decay is exponential, i.e., $KE_{\rm norm} \sim \exp(-\alpha N)$, where $\alpha = \alpha(w, q)$. For example, for $w = 0$, $\alpha = 0.05$. Since one may wonder why we did not study these variables over longer ranges, let us remind the reader that these systems are only of interest as small systems. It is not practical to experimentally construct these systems when they have, say, hundreds of grains. The grain sizes become large enough that the systems become unrealizable. Further, these small systems already exhibit rich dynamics, most of which is not analytically tractable, and large mass mismatches make even the simulations prone to integration errors.

To get a sense of how good an impact dispersion system the ITC is, we note that Fig. 6a suggests that $KE_{\rm norm}$ goes

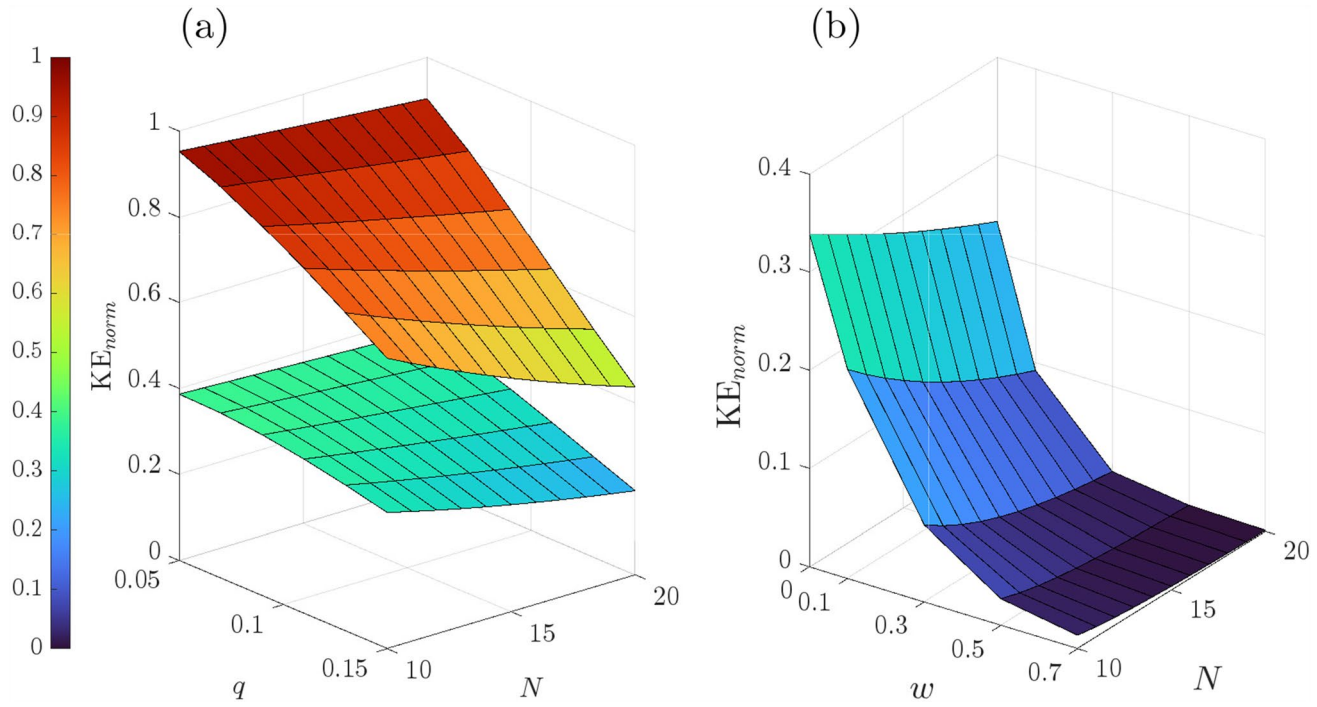


Fig. 6 In (**a**) the normalized kinetic energy is shown as a function of tapering (q) against the last grain in a system of N grains. In (**b**) the normalized kinetic energy is studied as a function of restitution w and system size N. The upper plot of (a) analytically models the hard sphere approximation per Eq. (8), whereas the lower plot models the actual system modeled via the Hertz potential soft-sphere and accounts for the potential energy that enters through the grain-grain potential. In (**b**), tapering q is set to 0.13 or 13%

down to ~ 0.25 or about 25% of the initial KE at $q = 0.1$ and $N = 20$ whereas for a TC $KE_{\rm norm}$ goes down to 0.1 or 10% of the original KE [38, 39]. However, the ITC is also a system in which the SW structure persists in the system, thus making it a combination of an impact dispersion system and a system in which a SW effectively ends up expanding as it travels.

The behavior of the $KE_{\rm norm}$ turns out to be sensitive to the restitution w and N for $q = 0.13$ as can be seen in Fig. 6b. For restitution $w = 0.20$, some 95% of the initial KE is lost in transit through the system. Of course, for such high restitution cases, the SW structure becomes too weak to be of relevance.

4 Conclusions

The present paper examines the propagation of a velocity impulse initiated at the smallest grain and into the chain in an ITC system. In our studies, the system is held in a frictionless container between rigid end walls. The grains are assumed to grow *linearly* from left to right. We focus on two key results in this work. First, we study how the perturbation travels into the chain. We find that the SW which forms out of the initial velocity impulse, progressively expands in width (due to the increasing widths of the grains) while carrying progressively less maximum kinetic energy (see Fig. 4). These waves actually lose energy even in energy conserved systems because there is energy backscattering at each granular contact. Thus, the dynamics of this system is rich and the dynamical equations are not amenable to an analytical solution. Second, we examine whether or not the ITC serves as an impact absorber. Here we find that the ITC is peculiar in the sense that it is a reasonably good impact absorber but it also has the simultaneous characteristic that the impulse travels as a SW which is ever widening. The fact that the energy travels as a SW implies that the system also transports the energy in a weakly dispersive manner, and in this context it is drastically different from the TC which is a versatile impact dispersion system.

The Hertz potential that facilitates the interactions at each grain-grain contact is softer than the harmonic potential [11] for small compressions. Thus, potential energy held at the grain-grain contacts as a function of time turns out to be an important piece in the dynamical problem. Our hard sphere approximation fails to capture this piece, which results in the dramatic difference in the values of $KE_{\rm norm}$ between the hard sphere and soft sphere cases in Fig. 6a. Nevertheless, the hard sphere approximation returns a reasonable description of how $KE_{\rm norm}$ behaves as functions of q and N as shown in Fig. 6.

Looking into the future, the ITC has opened the unexpected possibility that the SWs in these simple granular systems in principle can be made arbitrarily wide, with reduced linear energy densities, which in itself is a part of its impact damping. Novel ways of using the wider SWs may prove to be useful for practical applications.

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Author contributions The work was conceptualized by SS. The simulations and graphics work (using Matlab) were carried out by EB. The writing was done by SS in close collaboration with EB. Final edits were done by SS and EB.

Data availability The work is based on analyzing large amounts of dynamical simulation data that we have generated using our code. The data is already captured in the manuscript as embedded graphics. Additionally, we are happy to make the source code available to generate the data upon request to the corresponding author.

Declarations

Conflict of interest The authors declare that they have no Conflict of interest.

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